

mse/medse测试结果

代码部分

Loss function设置

```
def mse_loss(
    y_true: torch.Tensor,
    y_pred: torch.Tensor,
    reduction: Reduction = "mean",
) -> torch.Tensor:
    """
    Standard mean squared error loss used for NN_Mean.
    常规均方误差，用作 NN_Mean 的基准损失。
    """
    loss = (y_true - y_pred) ** 2
    return _reduce(loss, reduction)

def medse_loss(
    y_true: torch.Tensor,
    y_pred: torch.Tensor,
    reduction: Reduction = "median",
) -> torch.Tensor:
    """
    Median squared error for robustness against outliers.
    以平方误差的中位数为准的鲁棒损失，弱化异常值的影响。
    """
    loss = (y_true - y_pred) ** 2
    return _reduce(loss, reduction)
```

回测策略：long/short，rebalance设计

```
def compute_portfolio_returns(
    preds: np.ndarray, targets: np.ndarray
) -> Dict[str, float]:
    """
    Rebalance & Return 逻辑:
    - 每月按预测值排序, Top 10% 做多, Bottom 10% 做空。
    - 组合收益 = long mean - short mean.
    """
    n = preds.size # preds: 预测值
    if n == 0:
        return {"long": np.nan, "short": np.nan, "long_short": np.nan}
    order = np.argsort(preds)
    k = max(1, int(np.floor(n * 0.1)))
    long_idx = order[-k:]
    short_idx = order[:k]
    long_ret = float(np.mean(targets[long_idx])) if long_idx.size else float("nan")
    short_ret = float(np.mean(targets[short_idx])) if short_idx.size else float("nan")
    long_short = (
        float(long_ret - short_ret)
        if np.isfinite(long_ret) and np.isfinite(short_ret)
        else float("nan")
    )
    return {"long": long_ret, "short": short_ret, "long_short": long_short}
```

训练时长/窗口：选择1989.12-1994.12五年数据，仅预测1995.1-1995.6 六个月

预测结果

MSE:

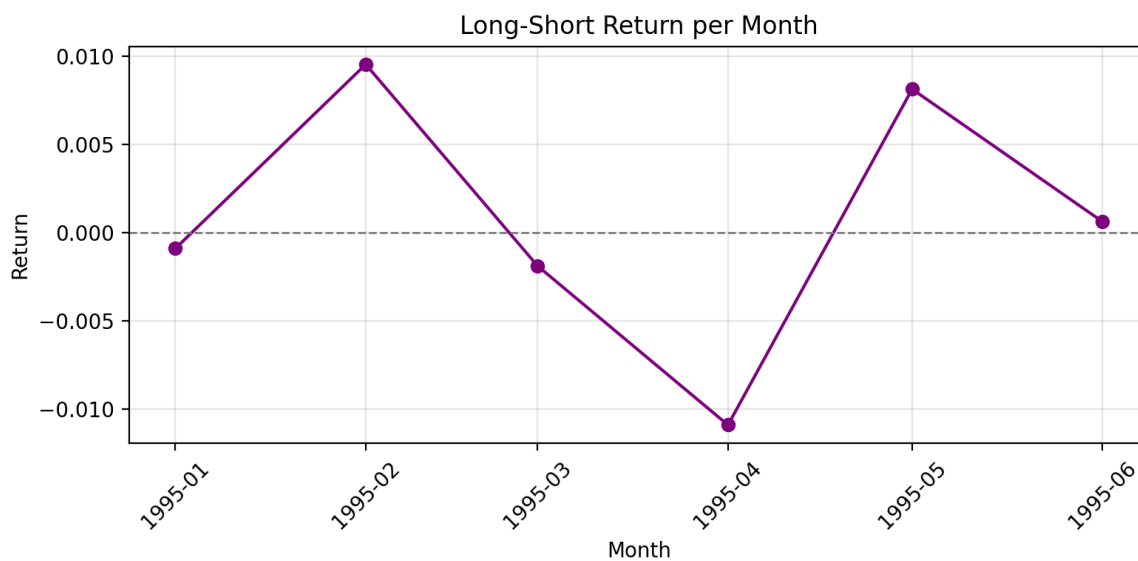
Rebalancing monthly: Top 10% predictions → Long, Bottom 10% → Short.

Month 1995-01	MSE 0.144110	R2 -2.2853	Long 0.0372	Short 0.0381	Long-Short -0.0009
Month 1995-02	MSE 0.083519	R2 -2.5766	Long 0.0303	Short 0.0207	Long-Short 0.0096
Month 1995-03	MSE 0.253002	R2 -9.0932	Long 0.0252	Short 0.0271	Long-Short -0.0019
Month 1995-04	MSE 0.098865	R2 -2.4386	Long 0.0263	Short 0.0372	Long-Short -0.0109
Month 1995-05	MSE 0.297344	R2 -8.4381	Long 0.0554	Short 0.0472	Long-Short 0.0081
Month 1995-06	MSE 1.174731	R2 -37.9274	Long 0.0593	Short 0.0586	Long-Short 0.0006

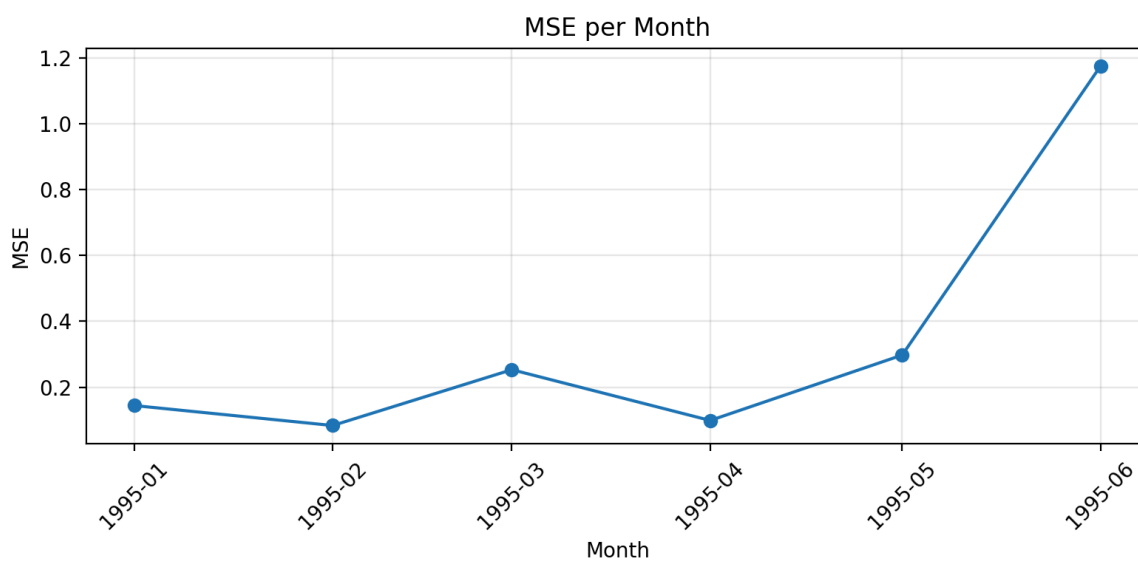
Saved metrics for MSE to sanity_outputs/sanity_metrics_mse.csv

Wrote summary to sanity_outputs/sanity_summary_mse.json
Loss curve saved to sanity_outputs/mse_loss_curve.png
Long-Short return curve saved to sanity_outputs/mse_returns_curve.png
MSE 实验完成。

return:



MSE 值



MEDSE

Rebalancing monthly: Top 10% predictions → Long, Bottom 10% → Short.

Month 1995-01 | MedSE 0.007704 | R2 -90338.9228 | Long 0.0237 | Short 0.0436 | Long-Short -0.0199

Month 1995-02 | MedSE 0.007293 | R2 -168889.7533 | Long 0.0140 | Short 0.0299 | Long-Short -0.0159

Month 1995-03 | MedSE 0.006875 | R2 -81002.7334 | Long 0.0259 | Short 0.0203 | Long-Short 0.0056

Month 1995-04 | MedSE 0.007301 | R2 -92210.7741 | Long 0.0215 | Short 0.0180 | Long-Short 0.0035

Month 1995-05 | MedSE 0.008806 | R2 -557850.1801 | Long 0.0465 | Short 0.0480 | Long-Short -0.0015

Month 1995-06 | MedSE 0.009170 | R2 -164280.4125 | Long 0.0498 | Short 0.0600 | Long-Short -0.0102

Saved metrics for MedSE to sanity_outputs/sanity_metrics_medse.csv

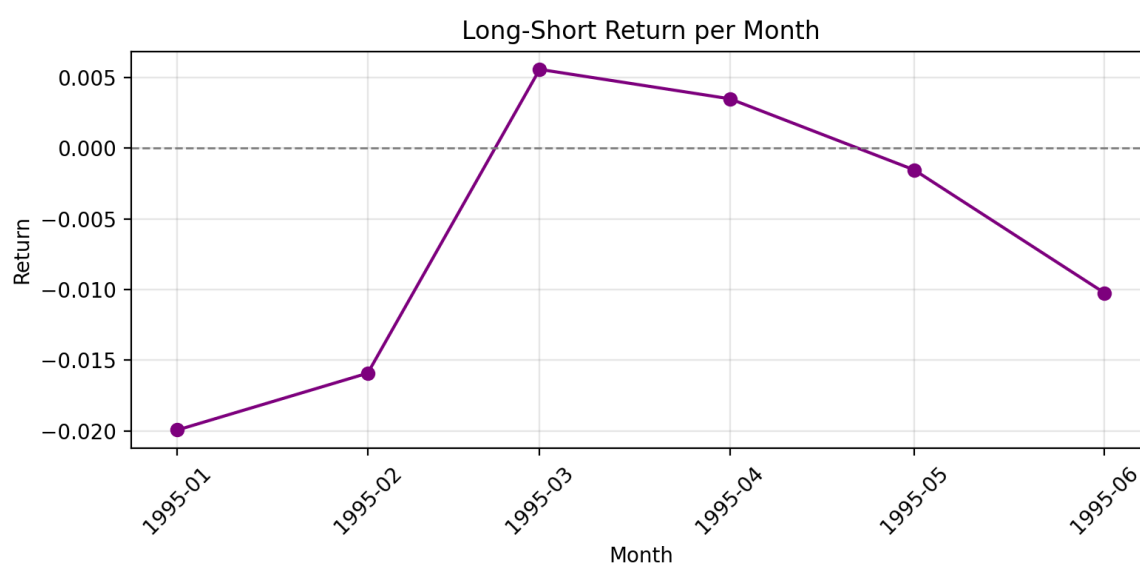
Wrote summary to sanity_outputs/sanity_summary_medse.json

Loss curve saved to sanity_outputs/medse_loss_curve.png

Long-Short return curve saved to sanity_outputs/medse_returns_curve.png

MedSE 实验完成。

return



Medse值

